

SHRESTH JAISWAL

Bhopal, India | +91 8470980674 | jaishresth143@gmail.com

[LinkedIn](#) | [GitHub](#) | [Portfolio](#) | [LeetCode](#)

EDUCATION

B.Tech in Computer Science and Engineering — Vellore Institute of Technology

Sep 2023 – Present

CGPA: 8.40 / 10.0

SKILLS

AI/ML & LLMs: LLM APIs (OpenAI GPT-4o, Gemini), Prompt Engineering, Conversational AI, Agentic Workflows (Function Calling, Tool Use), AI Output Evaluation

Voice AI: Pipecat, Deepgram Nova-2 (STT), ElevenLabs (TTS), Twilio Voice, WebRTC, Silero VAD

Languages: Java, Python, JavaScript, SQL

Backend: Django, FastAPI, Node.js, Express.js, Spring Boot, REST APIs, SQLAlchemy, Alembic

Databases: PostgreSQL, SQL

Frontend: React.js, Redux Toolkit, React Hook Form, HTML, CSS, TailwindCSS

Testing & Automation: Pytest, Selenium, API Testing, Unit Testing, Integration Testing

DevOps & Tools: Git, GitHub Actions, Docker, Linux

EXPERIENCE

Software Engineering Intern — Scramble Labs Technology Pvt. Ltd.

May 2025 – Aug 2025

- Engineered 2 production-ready REST APIs using Python, Django, and PostgreSQL, integrating request validation, database optimization, and RESTful architecture to support scalable backend services.
- Developed a Git workflow that standardized code merging for the IT department, preventing 15 critical merge conflicts and improving team collaboration.
- Integrated a comprehensive suite of 15 automated unit and integration tests for newly developed REST APIs.
- Developed 4 modular React.js + TypeScript components within a production Next.js application, reducing page load latency by 15% through React component caching and memoized render hooks.

PROJECTS

EstateStream AI — Real-Time Voice AI Agent for Real-Estate Sales

Python, FastAPI, Pipecat AI, OpenAI GPT-4o, Deepgram Nova-2, ElevenLabs, Twilio, PostgreSQL, Docker | [Code](#) | [Live Demo](#)

- Built and configured an inbound voice AI agent that qualifies real-estate leads over live phone calls, achieving <800ms end-to-end conversational latency by chaining streaming Deepgram Nova-2 (STT), GPT-4o, and ElevenLabs Turbo v2.5 (TTS) via Pipecat.
- Integrated the agent with Twilio Media Streams for live PSTN broker/customer calls over 8kHz μ -law WebSockets, plus a Web Audio API browser client with real-time waveform visualization.
- Designed conversation flows, guardrails, and agent personas, and implemented native GPT-4o function calling to ground responses in live property inventory and CRM data — eliminating hallucinated answers across all 19 tested call scenarios while persisting structured buyer profiles.
- Tested the agent across 19 automated scenarios spanning 10 adversarial failure modes (barge-in, long silences, hostile de-escalation), analyzing call logs to iteratively refine prompts and guardrails, and built an async SQLAlchemy/Alembic pipeline that auto-generates call summaries and outcome tags for the CRM.

TestPilot — AI-Powered Test Automation Platform

Python, FastAPI, Pytest, Selenium, PostgreSQL, LLM API, Docker | [Code](#) | [Live Demo](#)

- Designed structured prompts and engineered an AI-driven test generation pipeline that converts API specifications into executable test cases, covering functional, negative, and edge-case scenarios.
- Built an LLM evaluation framework to assess AI-generated test cases across correctness, consistency, coverage, and practical usability, enabling measurable quality scoring.
- Automated API and UI validation using Pytest and Selenium, with reusable test workflows for authentication, response schemas, status codes, and critical user journeys.
- Containerized and productionized the testing platform with Docker, PostgreSQL, and GitHub Actions for reproducible environments, automated test execution, and centralized test-result tracking.

TicketFlow — Enterprise IT Service Desk & Ticketing Platform

Java 17, Spring Boot 3, React 18, TailwindCSS, PostgreSQL, OpenAI GPT-4o / Gemini API | [Code](#) | [Live Demo](#)

- Architected a full-stack IT service desk platform with a Spring Boot 3 (Java 17) REST API and a React 18 + Vite + TailwindCSS frontend, with role-aware queues for employees, support agents, and admins.
- Integrated OpenAI GPT-4o/Gemini to auto-triage incoming tickets — classifying priority and department and drafting first-response replies — with a graceful fallback so ticket creation never fails if the LLM API is unreachable.
- Enforced IDOR protections and role-based authorization (RBAC) across JPA queries and API endpoints, plus stateless JWT authentication with BCrypt password hashing and rate-limited ticket submission.

ACHIEVEMENTS & LEADERSHIP

- Community Leadership: Core Member of the Notion Community at VIT Bhopal — organized tech talks and peer learning workshops.
- Professional Certifications: Earned NPTEL Silver Elite Certification in Cloud Computing & Internet of Things (IoT).